

PES UNIVERSITY
Department of Computer Science and Engineering

Lab 3: Architectural Style Justification

Customizable Subscription Box Scheduler — Problem #35

Architecture Selection

We chose the Microservices Architecture for the Customizable Subscription Box Scheduler system.

Architectural Choice

The system is structured as a set of small, independently deployable services — Subscription Manager, Billing & Payment Service, Fulfillment Manifest Generator, Catalog & Inventory Data Store, and Notification Service — each communicating through well-defined provided/required interfaces (e.g., `IProcessBilling`, `ICatalogQuery`), rather than as tightly coupled horizontal layers or a single centralized server.

Two Reasons for This Choice

- **Independent scaling for bursty workloads:** Billing runs and fulfillment-manifest generation both spike sharply around each monthly renewal date, while everyday box customization traffic stays comparatively steady. As separate services, the Billing & Payment Service and Fulfillment Manifest Generator can each be scaled out only when needed, without over-provisioning the customer-facing Subscriber Web/App Interface.
- **Fault isolation across critical vs. non-critical paths:** A failure or slowdown in a non-essential service, such as Notification Service, does not block core subscriber actions like customizing a box or pausing a billing cycle, since each service can fail independently without cascading through the system.

Security Advantage

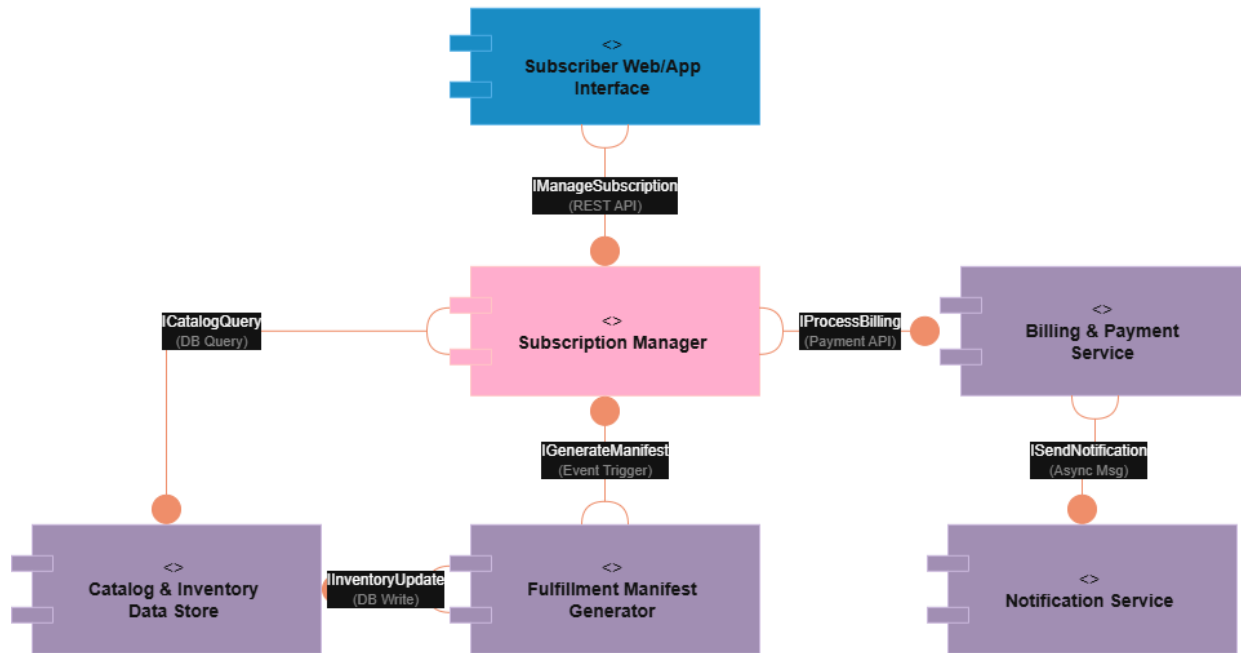
Isolating the Billing & Payment Service as its own microservice narrows the boundary that handles sensitive cardholder and billing data to a single, independently secured component. This reduces the attack surface: the Subscription Manager, Catalog & Inventory Data Store, and other services never need direct access to payment data, and access to the Billing & Payment Service can be locked down and audited separately from the rest of the system.

Performance Benefit

Because the Fulfillment Manifest Generator is deployed and scaled independently of the rest of the system, it can be allocated additional compute resources specifically during the monthly billing cycle to meet the requirement of exporting shipping labels for 10,000 boxes in under a minute, without that surge in demand degrading the responsiveness of the Subscriber Web/App Interface for customers who are simply browsing or customizing their box.

UML DIAGRAM

Component Diagram — Customizable Subscription Box Scheduler



Legend



Provided Interface (ball) — services this component offers

Required Interface (socket) — services this component needs



Component («component» stereotype)